# UsableNow

# Open Doesn't Mean Usable

Need-aware facility usability intelligence

**Lumina'26 — Hack the Unseen**

**Team Name:** 404 Brain Not Found

**Team Members:** Pragya Ahlawat (Team Leader)| Paras | Ronak Jain

Live Demo: https://ronak1167.github.io/UsableNow · GitHub: https://github.com/Ronak1167/UsableNow

PROBLEM STATEMENT

# The last mile of facility access still fails.

A facility listed as open does not guarantee a specific feature works right now.

Water point empty. Printer down. Laundry full. Lift unavailable. Step-free route blocked.

Wasted trips. Disrupted schedules. Avoidable effort and unequal access barriers.

**Which option is most likely to meet my need right now?**

This is the unanswered question.

TARGET USERS

# Built for the people who feel the friction.

## Who we serve

JIIT students & hostel residents (Shastri, Tagore, Sarojini) making time-sensitive trips across 17 campus facilities

People with mobility and accessibility needs who need reliable access

Campus visitors, students relying on shared services, and student organisers

## What they need

Current status of a facility or feature

When it was last checked

Source confidence they can trust

Better alternatives when plans change

**A typical moment**

01 → 02 → 03 → 04

Needs to use a facility before class

Checks status in UsableNow before leaving

Sees a recent unavailable report

Chooses a better available alternative

WHY IT MATTERS

# Why the problem is ‘unseen’

**Small failures** hide inside otherwise normal routines.

Open/closed labels and star ratings rarely describe **feature-level status.**

Conditions **change**; old reports can mislead.

A broken lift or inaccessible entrance imposes **disproportionate cost.**

Empirically validated across 17 facilities at JIIT-62 & hostels, tracking 9 distinct features with confidence decay and fail-closed report moderation.

PROPOSED SOLUTION

# A need-aware usability layer.
# From insight to action.

**1 Identify the need**

Select a need and choose the campus or area.

**2 Review evidence**

Inspect feature-specific reports with timestamps and source type.

**3 Compare options**

Evaluate choices by freshness, agreement, and walking routes.

**4 Submit update**

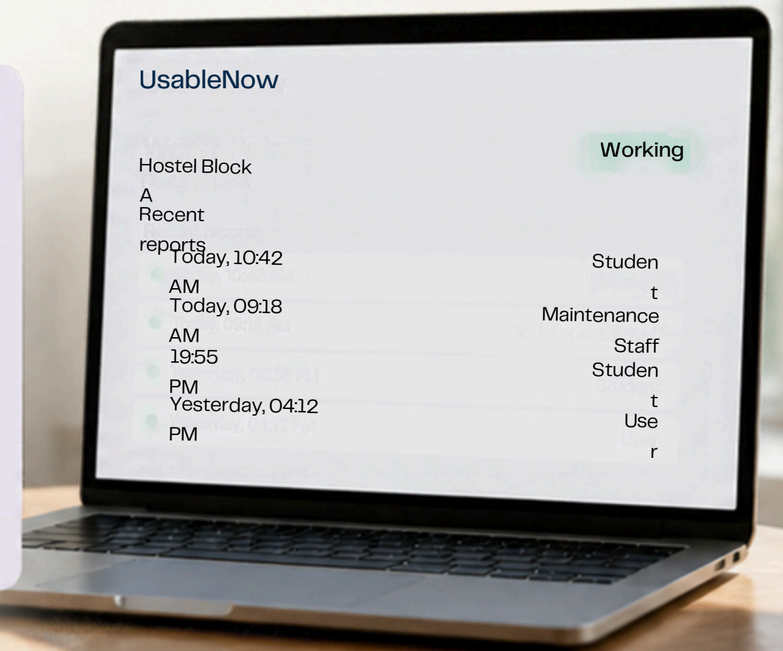
Submit reports with anti-spam rate limits & admin review queue.

RESULT

## Useful, actionable evidence.

Unknown is not the same as unavailable.

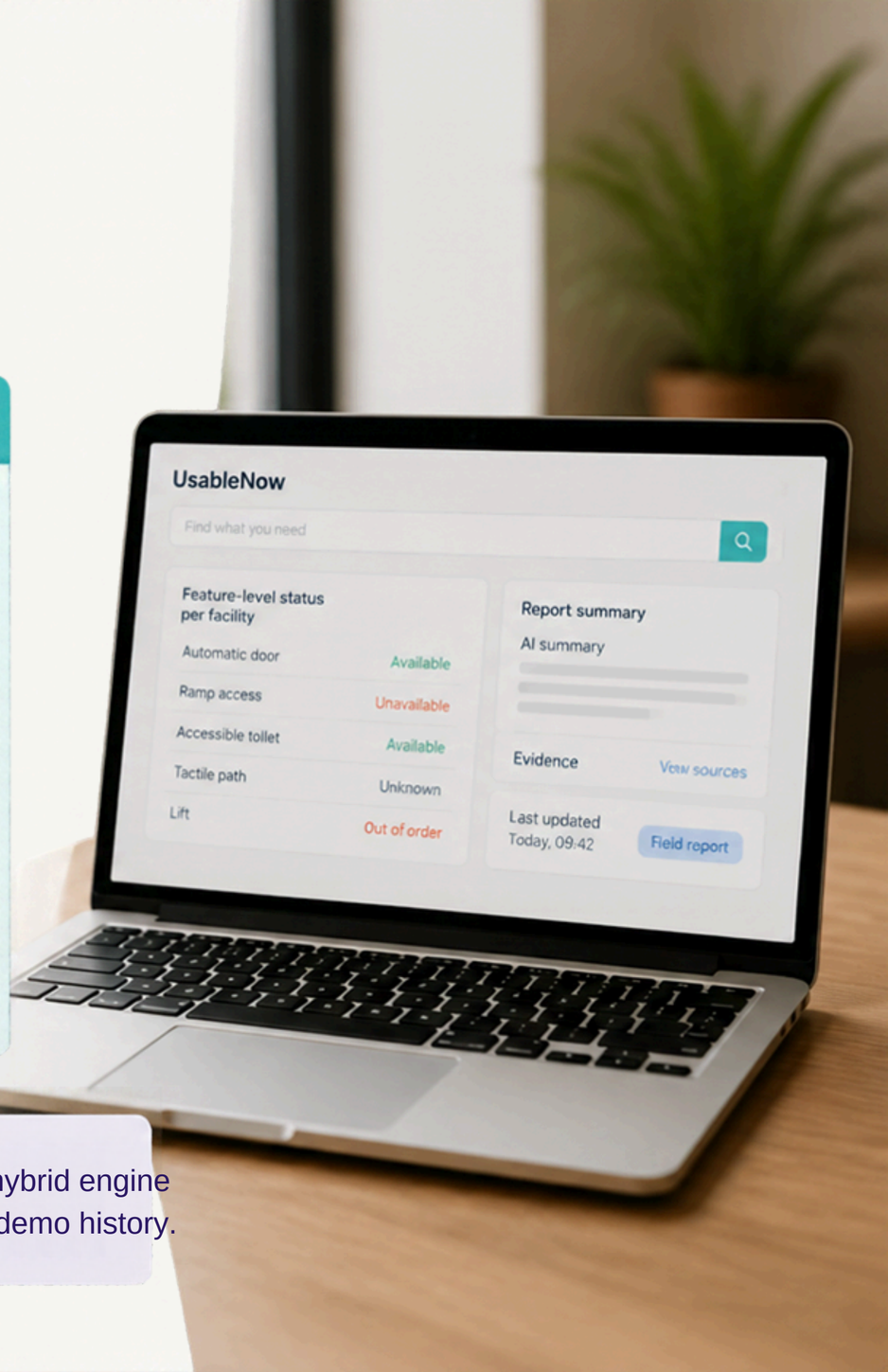
| | |
|---|---|
| Last checked | Today, 10:42 AM |
| Confidence | High |
| Why recommended | Multiple recent reports agree; matches your need in this area. |



# Key Features & Differentiation

| | |
|---|---|
| 01 | Need-based search not location-only search |
| 02 | Feature-level status per facility |
| 03 | Timestamped reports and source labels |
| 04 | Freshness decay and conflicting-report indicators |
| 05 | AI plain-language summaries with visible evidence |

| TYPICAL DISCOVERY | USABLENOW |
|---|---|
| Searches by location or name only | Searches by need, not just location |
| General accessibility status per facility | Feature-level status for every facility |
| Information often untimestamped | Every report is timestamped with source labels |
| Hard to tell what's outdated or conflicting | Shows freshness decay and conflicting reports |
| Manual review and stitching required | AI summary with visible evidence |

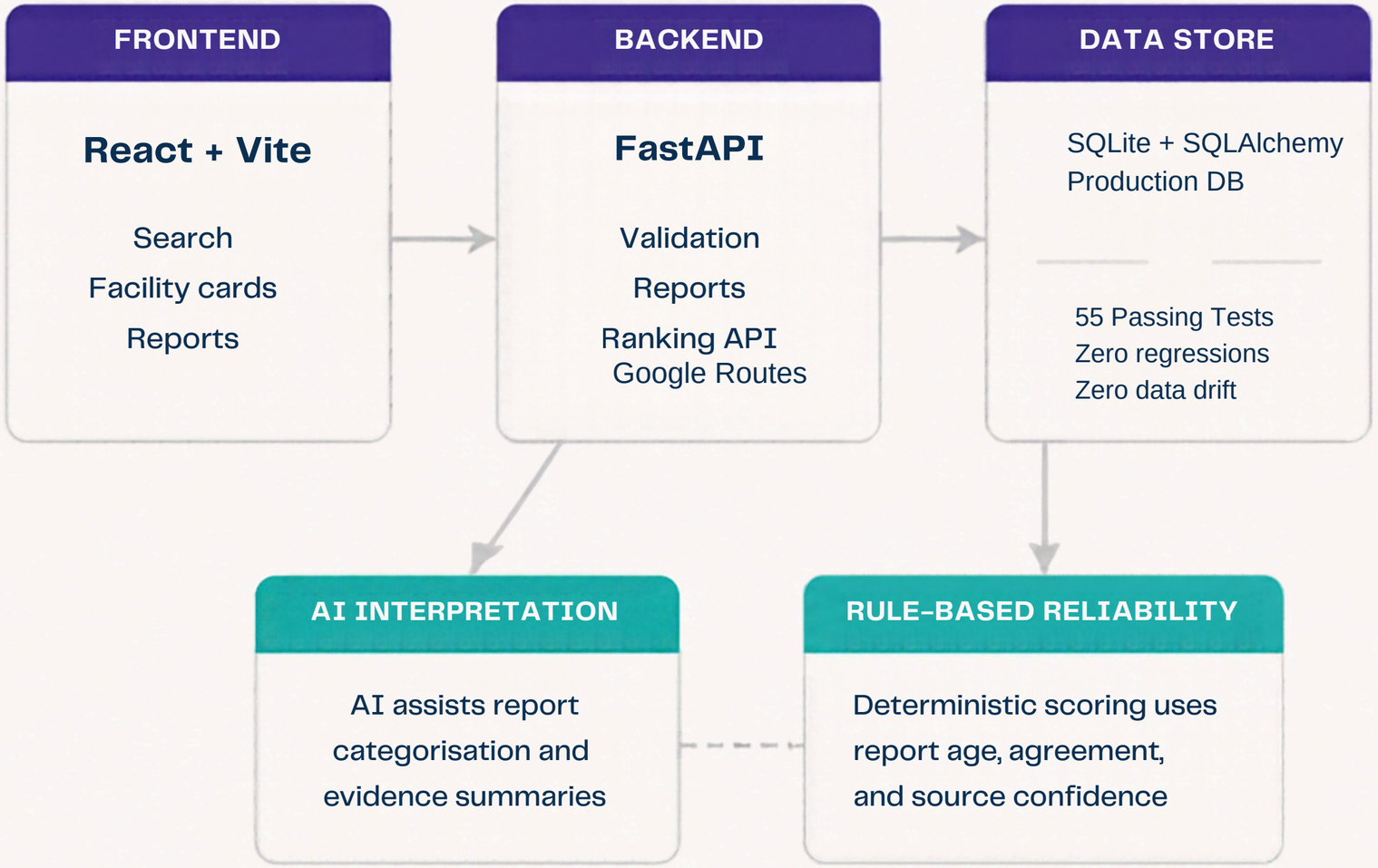



Delivers multi-feature independence across 17 campus facilities, real-time hostel laundry tracking, and Google Maps walking routes.

Fail-closed moderation & hybrid engine with live precedence over demo history.

# Technology & AI/ML Approach

Separation of AI interpretation and rule-based reliability.




## SAFETY BY DESIGN

**No custom-trained**
**ML model claimed**

Decision logic remains transparent and explainable.

**Minimise**
**personal data**

Collect only what is needed to improve usability.

**Discourage**
**spam and abuse**

IP submission rate-limiting, campus geofencing, and fail-closed admin secret validation.

**Show uncertainty**
**and expire stale reports**

Surface confidence levels and prioritise fresh, relevant reports.

# Current Progress,

## Delivered MVP & Empirical Verification

### WHAT'S CONFIRMED

- Problem student-validated
- Persona accessibility needs
- 9 usability feature models
- Production SQLite schema

### DELIVERED MVP

- 17 active campus facilities
- 3 hostel laundry room trackers
- Google Maps walking routes
- Admin moderation review queue

### EMPIRICAL PROOF

- 55 automated backend tests pass
- Zero frontend lint errors (oxlint)
- Production build clean (564ms)
- Live deployed on GitHub Pages

### VERIFIED METRICS

17 Facilities
Active & Mapped

9 Features
Real-Time Status

55 Tests
100% Pass Rate

0 Regressions
Production Ready

### SECURITY GATES

- IP submission rate limits (429)
- Fail-closed admin authentication
- Campus boundary geofence

### RELIABILITY ENGINE

- Live evidence takes precedence
- Exponential freshness decay
- Multi-report corroboration

### TEAM

**Pragya Ahlawat (Lead)**
**Paras**
**Ronak Jain**